HEALTHCARE ANALYTICAL PIPELINE

Data Analytics Portfolio Project

Tools Used -



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EXECUTIVE PROJECT INTRODUCTION

This end-to-end data enterprise assessment integrates financial, operational, and clinical systems across a healthcare delivery network encompassing **3,500 unique patient master index records** and **9,000 distinct inpatient encounters**.

Moving beyond isolated reporting, this portfolio project establishes a robust three-tier data pipeline:

1. **Data Ingestion & Integrity Layer (Excel / Power Query):** Resolves messy multi-format date records, normalizes missing payer variables, standardizes data types, and filters out 150 system anomalies.

1. **Relational Engineering & Query Layer (SQL):** Normalizes unrefined schemas into a structured Star Schema, writing production-grade queries, dynamic window functions, and subqueries to calculate hospital operational logic.

1. **Business Intelligence Visualization Layer (Power BI):** Deploys a native DAX modeling engine to build interactive dashboards featuring multi-page reporting, custom filtering panels, and proactive conditional logic.

Analytical & Strategic Scope

The pipeline tracks a fiscal ecosystem generating **$147.38M in gross charges** across six primary clinical specialties. By auditing structural operational variance—ranging from front-end intake integrity to 30-day post-discharge care transitions—this end-to-end roadmap provides executive leadership with actionable insights to maximize clinical team capacity, eliminate net collection leaks, and insulate the organization against costly clinical quality readmission penalties.

DATA CLEANSING & TRANSFORMATION LAYER (EXCEL & POWER QUERY)

1. Text & Structural Standardization

Applied TRIM, CLEAN, and PROPER text functions to normalize text variation across data rows.
Corrected mismatched column data types to enforce database storage compliance.
Removed 150 duplicate entries from the dataset using the native deduplication engine.

2. Destructive Format & Date Resolution

Messy Date Handling: Fixed *mixed text* formats *(D/M/Y, Y/M/D, M/D/Y, D-M-Y)* and unified them into a standard format (Y-M-D).

Date Engineering Execution: Leveraged Power Query features (Custom Columns, Column Splitting, and Merging) to successfully reconstruct clean date rows.

Financial Column Cleanup: Cleaned messy financial metrics featuring mixed strings of text, numeric fields, and embedded *"$"* formatting rules to ensure proper decimal types.

3. Data Value Imputation & Quality Rules

Abbreviated Text Normalization: Replaced ambiguous/shortened records with full structural names (e.g., M & F →Male & Female in the Patients table; Er →Emergency in the Admissions table; Y/Yes/N/No unified to true Yes & No.

Missing Insurance Profiles: Replaced blank spaces inside the “Insurance_Provider” column with the clear string designation "N/A“ to prevent system processing leaks.

Statistical Imputation: - Replaced missing cells in the “Patient_Satisfaction” column with the calculated statistical median rating.

- *Corrected corrupted Age metrics* containing *blanks, -1, and 999* by substituting them with the cohort’s dynamic *median age.*

4. Calculated Custom Columns

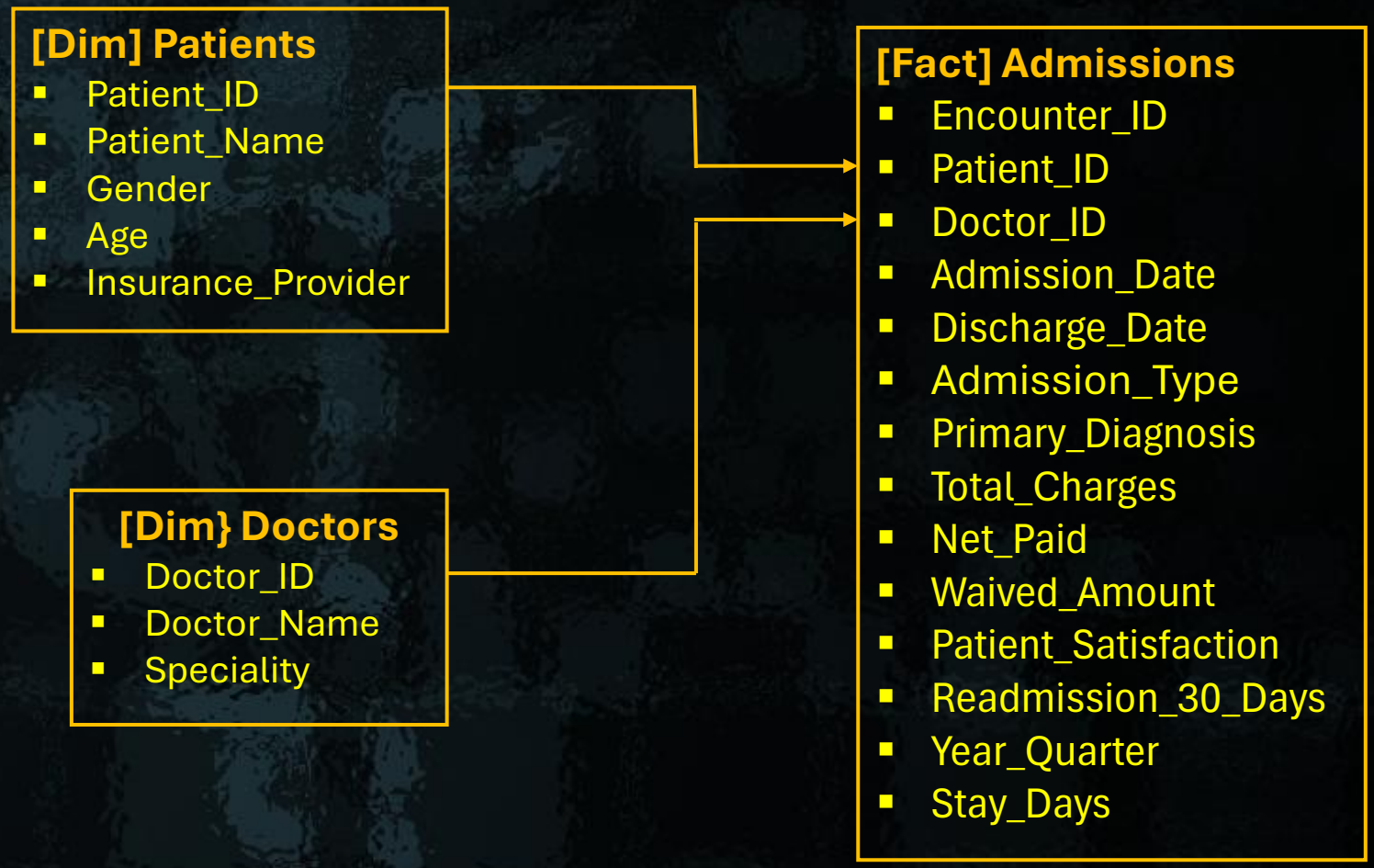
Waived_Amount: Created a new calculated column by subtracting the dynamic “Net_Paid” fields from “Total_Charges”.

Year_Quarter: Extracted dynamic calendar periods straight from the clean “Admission_Date” value.

Stay_Days: Formulated an emergency department metric to calculate total length of stay (LOS) variations across distinct patient lines.

RELATIONAL ENGINEERING & QUERY LAYER (SQL)

1. **Database Architecture & Optimization**
   - Imported clean Excel datasets directly into the database engine.
   - Established relational integrity using explicit Primary Key (PK) and Foreign Key (FK) table constraints.



2. Applied Analytical Commands

Data Filtration & Slicing: Utilized structured relational operations (SELECT, WHERE, HAVING, DISTINCT) to parse historical admission blocks.

Mathematical Aggregation Layer: Deployed advanced calculations (SUM, AVG, COUNT, MIN, MAX) to summarize top-tier hospital metrics.

Data Set Junctions: Written comprehensive row joins (INNER JOIN, LEFT JOIN, RIGHT JOIN) to connect multidimensional records smoothly across fact tables.

3. Production-Grade Query Frameworks

Logical Case Branching: Handled categorical field bucketing natively inside SQL scripts via CASE WHEN conditional mapping logic.

Advanced Analytics Engine: Designed highly performant CTEs (Common Table Expressions) and window functions (PARTITION BY, RANK(), ORDER BY) to deliver doctor performance insights and top-tier provider rankings.

Data Consolidation Architecture: Handled complex query result sets by structuring clean combinations using logical UNION ALL statements to form a final dashboard staging layer.

TOTAL PATIENT REGISTRY

```
-- total_patient in hospital record
select count(patient_id) as total_patients from patients;
```

total_patients
3500

Insight

The facility master patient index contains a baseline pool of *3,500 unique patient records*.

INPATIENT UTILIZATION

```sql
-- total_patients ever admited in the year irrespective of readmission
select count(distinct patient_id) as net_admited_patients_count from admissions;
```

net_admited_patients_count
3260

Insight

A total of *3,260 unique patients* generated active inpatient *admissions* over the fiscal year.

OUTPATIENT / PRE-ADMISSION DRIFT

```sql
-- patient count who never get admited but are in records.
select count(p.patient_id) from patients p left join admissions a
on p.patient_id=a.patient_id where a.patient_id is null ;
```

non_admission_patients
240

Insight

There are *240 registered individuals* who have *no recorded inpatient encounters*. These records represent pre-admission entries, outpatient clinic referrals, or data anomalies requiring automated front-end cleanup.

REVENUE

```sql
-- revenue
select sum(total_charges) as gross_revenue,sum(net_paid) as net_revenue,
sum(waived_amount) as waived_revenue from admissions;
```

gross_revenue	net_revenue	waived_revenue
147383471.51	115895409.34	31488062.17

Insight

Out of $147.38M in gross chargemaster revenue, $31.48M (21.3%) was accounted for as contractual adjustments, yielding a net allowed revenue of $115.89M. This falls within healthy industry baselines for negotiated payer write-offs.

CASE MIX VARIANCES

```
-- avg amounts
select round(avg(total_charges),2) as avg_bill,
round(avg(net_paid),2) as avg_paid_amount,
round(avg(waived_amount),2) as avg_waived_amount
 from admissions;
```

avg_bill	avg_paid_amount	avg_waived_amount
16375.94	12877.27	3498.67

Insight

The average gross charge per encounter stands at *$16,375.94*, with a net paid average of *$12,877.27.*

BILLING SYSTEM OUTLIERS

```sql
-- max and min amount
select min(total_charges) as min_bill ,max(total_charges) as max_bill,
min(net_paid) as min_paid_amount ,max(net_paid) as max_paid_amount,
min(waived_amount) as min_waived_amount,max(waived_amount) as max_waived_amount
 from admissions;
```

min_bill	max_bill	min_paid_amount	max_paid_amount	min_waived_amount	max_waived_amount
9279.65	30407.68	0.00	24740.07	8.26	21236.27

Insight

Minimum *write-offs* as *low as $8.26* indicate an opportunity to *implement automated billing thresholds* to reduce administrative claims processing overhead. Conversely, encounters reflecting *zero net collections* indicate charity care allocations, full payer denials, or total administrative write-offs that require automated secondary validation rules.

PAYER DISTRIBUTION

```sql
-- insurance provider wise
select p.insurance_provider,count(distinct p.patient_id) as patient_count,sum(a.total_charges) as gross_revenue,sum(a.net_paid) as net_revenue,
sum(waived_amount) as waived_revenue from patients p left join admissions a on p.patient_id = a.patient_id
group by insurance_provider order by gross_revenue desc;
```

insurance_provider	patient_count	gross_revenue	net_revenue	waived_revenue
Medicare	1019	42894461.15	33790081.68	9104379.47
Blue Cross	705	30016912.42	23787164.73	6229747.69
Medicaid	560	24239669.62	18951804.77	5287864.85
Cigna	353	14891247.66	11583830.46	3307417.20
Aetna	361	14115815.48	11246870.79	2868944.69
Self-Pay	317	13409707.09	10333144.76	3076562.33
N/A	185	7815658.09	6202512.15	1613145.94

Insights

Payer Concentration: *Medicare* commands the hospital's primary volume, driving **29%** of the total patient base and yielding the highest gross revenue collections. Combined with Blue Cross and Medicaid, the *top three payers* account for *65%* of the hospital's entire patient footprint.

Net Collection Rate:

Aetna	*79.68%*
N/A	*79.36%*
Blue Cross	*79.25%*
Medicare	*78.77%*
Medicaid	*78.19%*
Cigna	*77.79%*
Self-Pay	*77.06%*

Unassigned Insurance Vulnerability: Encounters categorized under missing insurance data *("N/A")* represent an operational vulnerability, accumulating *$1.6M* in unallocated adjustments. This underscores a front-end registration or demographic verification gap that directly compromises revenue capture.

Contractual Variance Drift: Despite maintaining nearly identical gross revenue profiles to Aetna, Cigna accounts for approximately $500K more in contractual allowances. This signal requires an active audit of the current Cigna fee schedule.

DEPARTMENT REVENUE

```sql
-- department wise revenue
select d.specialty as department,sum(a.total_charges) as gross_revenue,sum(a.net_paid) as net_revenue,
sum(a.waived_amount) as waived_revenue from doctors d
left join admissions a on d.doctor_id=a.doctor_id group by d.specialty order by gross_revenue desc;
```

department	gross_revenue	net_revenue	waived_revenue
General Surgery	29719191.55	23367706.26	6351485.29
Internal Medicine	29512064.37	23210220.49	6301843.88
Neurology	29143352.30	23025462.38	6117889.92
Pulmonology	23804285.04	18575182.61	5229102.43
Orthopedics	23296223.02	18378979.71	4917243.31
Cardiology	11908355.23	9337857.89	2570497.34

Insight

Cardiology and orthopaedics operate at the bottom of facility revenue generation, with *Cardiology pulling in less than half the volume of Orthopaedics*. Because both departments traditionally serve as high-margin, top-tier revenue drivers across the healthcare sector, their *poor performance* reflects *severe* local market *under-penetration.*

DEPARTMENT VS. AVERAGE FEE

```
-- department wise avg amounts
select d.specialty as department,round(avg(a.total_charges),2) as avg_bill,round(avg(a.net_paid),2) as avg_paid_amount,
round(avg(a.waived_amount),2) as avg_waived_amount from doctors d
left join admissions a on d.doctor_id=a.doctor_id group by d.specialty order by avg_bill desc;
```

department	avg_bill	avg_paid_amount	avg_waived_amount
Cardiology	16402.69	12862.06	3540.63
General Surgery	16401.32	12896.09	3505.23
Pulmonology	16382.85	12784.02	3598.83
Internal Medicine	16377.39	12880.26	3497.14
Neurology	16372.67	12935.65	3437.02
Orthopedics	16325.31	12879.45	3445.86

Insight

The average gross charge per patient case remains flat at *roughly $16.3K across every single department*. This uniform charge distribution implies that the low total revenue observed in Cardiology is driven entirely by a low patient encounter volume rather than a deficient pricing strategy.

DOCTOR PRODUCTIVITY AND CAPACITY

```sql
-- department wise doctor count and patient doctor ratio
select d.specialty as department,count(distinct d.doctor_id) as num_of_doctors,
round(count(a.encounter_id )/count( distinct d.doctor_id ),0) as patients_per_doctor
from doctors d left join admissions a on d.doctor_id = a.doctor_id group by department order by num_of_doctors desc ;
```

department	num_of_doctors	patients_per_doctor
General Surgery	10	181
Internal Medicine	10	180
Neurology	10	178
Orthopedics	8	178
Pulmonology	8	182
Cardiology	4	182

Insight

The facility averages roughly *180 unique patient* encounters *per doctor* annually across all departments. When translated to working shifts, this represents *a Daily Active* Census of approximately *4 patients per physician*. This *falls significantly below the standard inpatient* hospitalist workload threshold of *12 to 15 active patients per day*, proving the *medical staff is underutilized* and has immediate *structural capacity to scale up volume*.

TOP PERFORMING DOCTORS

```sql
-- best performing doctors for every department
with performance_ranking as(
select d.doctor_id,d.doctor_name,d.specialty as department,count(a.encounter_id)as patient_count,round(avg(a.patient_satisfaction),2) as satisfaction_rating,
rank() over(partition by d.specialty order by(count(a.encounter_id)) desc,round(avg(a.patient_satisfaction),2)desc)as rnk
from doctors d left join admissions a on d.doctor_id = a.doctor_id group by d.doctor_id,d.doctor_name,d.specialty
)
select doctor_id,doctor_name,department,patient_count,satisfaction_rating from performance_ranking
where rnk = 1 order by patient_count desc;
```

doctor_id	doctor_name	department	patient_count	satisfaction_rating
16	Dr. Charles Miller	Internal Medicine	209	3.56
32	Dr. Alex Smith	General Surgery	207	3.68
23	Dr. Mary Brown	Orthopedics	204	3.62
22	Dr. Linda Lopez	Pulmonology	203	3.68
1	Dr. David Martin	Neurology	193	3.67
28	Dr. Charles Gonzalez	Cardiology	190	3.64

Insight

These six represent the top-performing doctor within each of their respective department based on patient volume and satisfaction rating, led by Dr. Charles Miller (Internal Medicine, 209 cases) which is roughly 30 more patients than average(180).

The *satisfaction rating* averages *3.64 out of 5* among these departments top performers. This serves as an indicator of consistent baseline patient experience across the facility.

DEMOGRAPHIC PROFILE

```sql
-- patient count by age group
SELECT COUNT(CASE WHEN p.age BETWEEN 18 AND 35 THEN a.encounter_id END) AS young_18_to_35,
COUNT(CASE WHEN p.age BETWEEN 36 AND 65 THEN  a.encounter_id END) AS middle_aged_36_65,
COUNT(CASE WHEN p.age > 65 THEN  a.encounter_id END) AS senior_citizen_over_65
FROM patients p right join admissions a on p.patient_id = a.patient_id;
```

young_18_to_35	middle_aged_36_65	senior_citizen_over_65
824	5401	2775

Insight

Patients aged *35 and older* account for *over 91%* of total hospital volume, with middle-aged adults (36–65) serving as the facility's core consumer engine at 60% and seniors at nearly 31% . Strategic marketing initiatives, service-line planning, and clinical pathways should focus on chronic disease management and preventative care models optimized for this demographic.

DEPARTMENT DEMOGRAPHIC PROFILE

```
-- patients_count depatment wise and their avg_satisfaction rating(out of 5)
select d.specialty as department,count(a.encounter_id) as total_admissions,round(avg(a.patient_satisfaction),2) as avg_staisfaction_rating
 from doctors d left join admissions a on d.doctor_id=a.doctor_id group by department order by total_admissions desc ;
```

department	total_admissions	avg_staisfaction_rating
General Surgery	1812	3.61
Internal Medicine	1802	3.58
Neurology	1780	3.61
Pulmonology	1453	3.63
Orthopedics	1427	3.60
Cardiology	726	3.65

Insight

General Surgery and Internal Medicine have the highest patient volume with over 1800 each. Combined with Neurology, the patient volume of *the top 3 department* volumes nearly 5400 representing *approximately 60%* of the annual total of 9000 patients.

While consistent, a facility-wide satisfaction rating of *3.6 out of 5* places us firmly at industry average. Because this performance deficit spans all six departments equally, remedial strategies must target centralized systemetic processes.

ACQUISITION AND SATISFACTION CEILING

```sql
-- no.of patients by visit_count and their avg satisfaction rating(out of 5)
select visit_count,count(patient_id) as patient_count,round(avg(patient_satisfaction),2)as satisfaction_rating from
(select patient_id,count(encounter_id) as visit_count,avg(patient_satisfaction)as patient_satisfaction from admissions
group by patient_id) as t1 group by visit_count order by patient_count desc;
```

visit_count	patient_count	satisfaction_rating
2	853	3.63
3	777	3.61
1	725	3.57
4	509	3.59
5	248	3.63
6	97	3.67
7	36	3.51
8	10	3.61
9	3	3.52
10	1	3.50
11	1	4.00

Insight

Single-visit patients lag behind the volumes established by returning patients (2 to 3 visits). This shows that our facility's attraction of new, distinct patients into pipeline is low.

Patient satisfaction remains entirely *mediocre* at 3.6 out of 5, completely independent of visit frequency. The fact that returning patients report the same average experience as first-time visitors proves a systemic failure to deliver escalating value or seamless continuity of care.

Mediocre satisfaction among long-term patients eliminates advocacy, directly resulting in *lost* new patient acquisition *opportunities* that should be driven by positive word-of-mouth and *patient recommendations*.

AVERAGE LENGTH OF STAY (ALOS) INVARIANCE

```
-- avg_stay days
select primary_diagnosis,avg(stay_days) as avg_stay_days from admissions group  by primary_diagnosis ;
```

primary_diagnosis	avg_stay_days
Cholecystitis	4.0182
Sepsis	4.0216
Osteoarthritis	3.8679
Copd Exacerbation	4.0312
Appendicitis	4.0685
Diabetes Complications	3.9736
Urinary Tract Infection	4.1012
Hernia Repair	4.0791
Transient Ischemic Attack	4.0034
Pneumonia	3.9900
Severe Asthma	4.0794
Knee Replacement	4.0106
Hip Fracture	4.0000
Heart Failure	3.8008
Seizure Disorder	3.9632
Arrhythmia	4.0165
Ischemic Stroke	4.0621
Myocardial Infarction	4.0429

Insight

The facility's *overall ALOS* stands flat at 4.01 days. This uniformity persists across all primary diagnoses, indicating a *fixed discharge* protocol or standardized case management constraints *that override diagnostic severity variations*.

30-DAY READMISSION EXPOSURES

```sql
-- readmission within 30 days
select primary_diagnosis ,
round(count(case when readmission_30_days='yes' then 1 end)/count(readmission_30_days)*100,2) as within_30_days_readmission_rate
from admissions group by primary_diagnosis order by within_30_days_readmission_rate desc;
```

primary_diagnosis	within_30_days_readmission_rate
Urinary Tract Infection	21.92
Heart Failure	21.91
Diabetes Complications	20.30
Transient Ischemic Attack	19.97
Arrhythmia	19.83
Copd Exacerbation	19.30
Severe Asthma	19.27
Sepsis	18.91
Hip Fracture	18.02
Appendicitis	17.78
Pneumonia	17.64
Hernia Repair	17.34
Knee Replacement	17.16
Cholecystitis	16.53
Osteoarthritis	16.17
Ischemic Stroke	15.43
Seizure Disorder	15.36
Myocardial Infarction	14.16

avg_within_30_days_readmission_rate
18.14

Insight

The facility suffers from an *elevated average 30-day readmission rate* of *18.14%*. High-acuity conditions such as Urinary Tract Infections (21.92%) and Heart Failure (21.91%) present severe exposure under CMS quality penalty programs. Even the lowest readmission tier—Myocardial Infarction at 14.16%—remains a clear clinical risk.

PROVIDER-LEVEL QUALITY VARIANCE

```sql
-- doctor wise readmission within 30 days
with dw_re30 as(
select d.doctor_id,d.doctor_name,round(count(case when a.readmission_30_days='yes' then 1 end)/count(*)*100,2) as within_30_days_readmission_rate
from doctors d inner join admissions a on d.doctor_id = a.doctor_id group by d.doctor_id,d.doctor_name order by within_30_days_readmission_rate desc
)
(select 'highest_readmission_rate_doctors' as doctor_id,'' as doctor_name,'' as within_30_days_readmission_rate )
union all
(select * from dw_re30 order by within_30_days_readmission_rate desc limit 3)
union all
(select 'lowest_readmission_rate_doctors','' ,'')
union all
(select * from dw_re30 order by within_30_days_readmission_rate asc limit 3);
```

doctor_id	doctor_name	within_30_days_readmission_rate
highest_readmission_rate_doctors		
19	Dr. Linda Davis	26.14
10	Dr. Charles Jackson	24.10
17	Dr. Jessica Anderson	24.00
lowest_readmission_rate_doctors		
21	Dr. Mary Brown	12.63
4	Dr. Robert Brown	12.72
41	Dr. Jennifer Miller	12.90

Insight

Dr. Linda Davis exhibits the facility's highest individual 30-day readmission correlation at *26.14%*, whereas Dr. Mary Brown maintains the strongest clinical discharge stability with a low readmission rate of *12.63%*.

THE POST-DISCHARGE CARE GAP

```
-- stay_days vs readmission vs satisfaction rating
select distinct stay_days,round(count(case when readmission_30_days='yes' then 1 end)/count(*)*100,2) as within_30_days_readmission_rate,
avg(patient_satisfaction) as satisfaction_rating from admissions group by stay_days order by stay_days;
```

stay_days	within_30_days_readmission_rate	satisfaction_rating
1	19.20	3.5910
2	18.75	3.5575
3	17.51	3.5979
4	16.95	3.5853
5	17.77	3.6842
6	18.15	3.5961
7	21.88	3.6758
8	17.60	3.6440
9	20.83	3.6250
10	17.54	3.6316
11	21.43	3.4286
12	25.00	4.2500
13	0.00	4.0000
14	0.00	2.3333
17	0.00	3.0000

Insight

Inpatient readmission rate steadily *falls* from *Day 1 (19.20%) through Day 4 (16.95%),* but *spikes sharply* to a peak of *25.00%* for patients with a 12-day length of stay. This correlation proves that high-acuity, longer-stay patients *lack sufficient post-discharge stabilization*. Implementing a formal Transitional Care Management (TCM) framework or post-discharge telephone protocols within 48 hours of release is highly recommended to bridge this gap

Note: Rows after 12th day are volatile outliers because of low data volume.

BUSINESS INTELLIGENCE & DAX MODELING ENGINE (POWER BI)

1. **Relational Star Schema Implementation**
   - Imported the normalized, query-validated tables directly from the relational SQL database layer.
   - Established an optimized analytical Star Schema by mapping clean relationships across key fields:
     - Fact_Admissions →Linked to Dim_Patients via an active relationship on Patient_ID.
     - Fact_Admissions →Linked to Dim_Doctors via an active relationship on Doctor_ID.
   - Configured all relationship dimensions to enforce strict downstream data modeling standards and allow smooth cross-filtering across visuals.

2. **Row-Level Calculated Column Logic**

   Satisfaction_Breakdown: Deployed a logical SWITCH(TRUE()) structural evaluation rule to group raw survey scores into clean, meaningful text blocks for business tracking:

```
Satisfaction_Breakdown = SWITCH(TRUE(),Fact_Admissions[Patient_Satisfaction] >= 4,"Good",Fact_Admissions[Patient_Satisfaction] >=3,"Average","Poor")
```

3. **Production-Grade KPI Summary Measures**

   Base Volume Row Sizing: Implemented an optimized table counter to drive absolute throughput metrics:

```
row count = COUNTROWS(Fact_Admissions)
```

Net Revenue Conversion Efficiency: Formulated a calculation to evaluate overall financial health by tracking net realized collections against gross billings:

```
Net_Conversion_Rate = DIVIDE(SUM(Fact_Admissions[Net_Paid]),SUM(Fact_Admissions[Total_Charges]))
```

Case Acuity Performance Metrics: Tracked standard hospital stay variations by generating a precise decimal average length of stay (ALOS):

```
ALOS = ROUND(AVERAGE(Fact_Admissions[Stay_Days]),2) & " Days"
```

4. **Operational Throughput Velocity Formulas**

Daily Admission Flux: Calculated the baseline incoming patient velocity over a standardized annual calendar period

```
Avg. Admissions = ROUND(CALCULATE(COUNTROWS(Fact_Admissions))/DISTINCTCOUNT(Fact_Admissions[Admission_Date]),0) & "/" & "Day"
```

Dynamic Discharge Velocity: Structured a filter framework restricted to the 2025 fiscal data boundaries to evaluate outward team capacity:

```
Avg.Discharge = ROUND(CALCULATE(COUNTROWS(Fact_Admissions),FILTER(Fact_Admissions,YEAR(Fact_Admissions[Discharge_Date])= 2025 ))/CALCULATE
(DISTINCTCOUNT(Fact_Admissions[Discharge_Date]),FILTER(Fact_Admissions,YEAR(Fact_Admissions[Discharge_Date]) = 2025)),0) & "/" & "Day"
```

5. **Payer Gaps & Value-Based Purchasing Penalties**

Zero-Collection Adjustments: Engineered an audit metric to instantly isolate the exact count of completely uncollected patient encounters:

```
total no. of fully waived bill = CALCULATE(COUNTROWS(Fact_Admissions),FILTER(Fact_Admissions,Fact_Admissions[Net_Paid] = 0))
```

CMS Quality Penalty Exposure Rate: Formulated an aggregate risk measure to track the dynamic percentage of historical 30-day clinical readmissions:

```
Total Readmited Cases = CALCULATE(COUNTROWS(Fact_Admissions),FILTER(Fact_Admissions,Fact_Admissions[Readmission_30_Days]="Yes"))
```

```
Readmission % 30 days = CALCULATE(COUNTROWS(Fact_Admissions),filter(Fact_Admissions,Fact_Admissions[Readmission_30_Days] = "Yes"))/[row count]
```

DASHBOARD – LANDING & PROJECT HOME

Healthcare Project

Revenue Cycle Management (RCM) & Financial Performance

Evaluates the hospital's baseline financial health by monitoring the gap between gross billings and actual cash collections. This dashboard tracks net collection rates, audits payer contract compliance, and uncovers unallocated revenue leakages.

Patient Demographics & Patient Intake Integrity

Analyzes consumer volume engines across departments, insurance provider age cohorts and gender distributions to optimize clinical service line capacity. Tracks chronological admission trends and assesses intake acuity mixes to align staffing resources.

Clinical Quality & Satisfaction Ceiling

Audits clinical discharge stability against CMS readmission guidelines to protect the network from financial penalties. Cross-references Average Length of Stay (ALOS) with patient satisfaction marks to eliminate facility-wide service constraints.

Note: **This report is based on data of admissions from 1 January 2025 till 31 December 2025.**



❑ Navigation Framework & Landing Logic

Executive Navigation Architecture: Designed a unified modular landing hub to direct corporate leadership immediately to the three primary analytics categories: Financials, Demographics, and Clinical Quality.

Timeframe Boundaries: Explicitly documented data parameters on the landing screen to isolate the exact timeline limits of the underlying model, bounding the data between **January 1, 2025, and December 31, 2025**.

Systemic Overview Blocks: Structured descriptive summary buckets at the baseline entry point to explain the exact operational objectives of each tracking view before users drill into deeper pages.

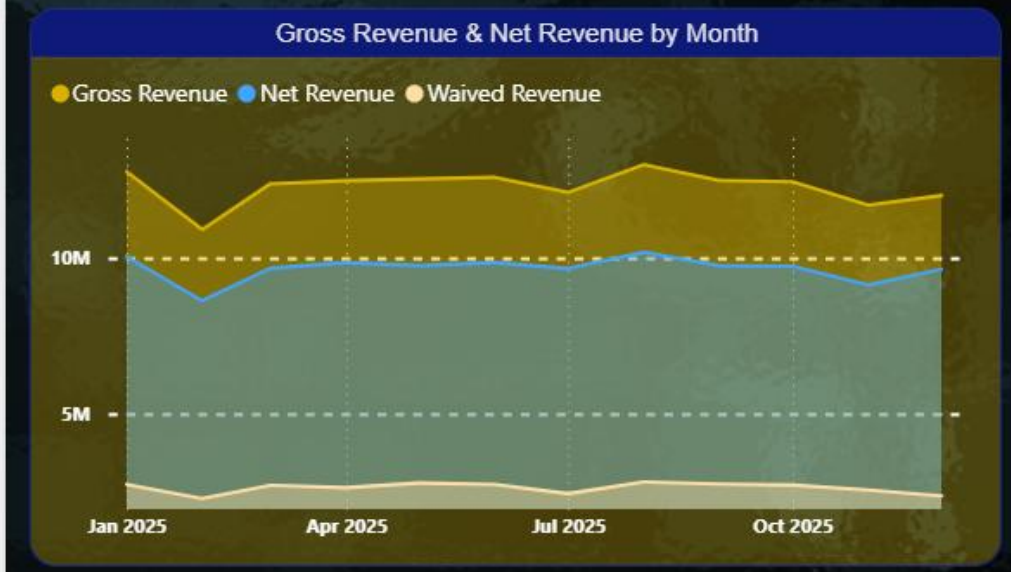
Visual Continuity: Established a dark, modern aesthetic with color-coded theme navigation indicators that repeat across all sub-pages to ensure a cohesive user experience.

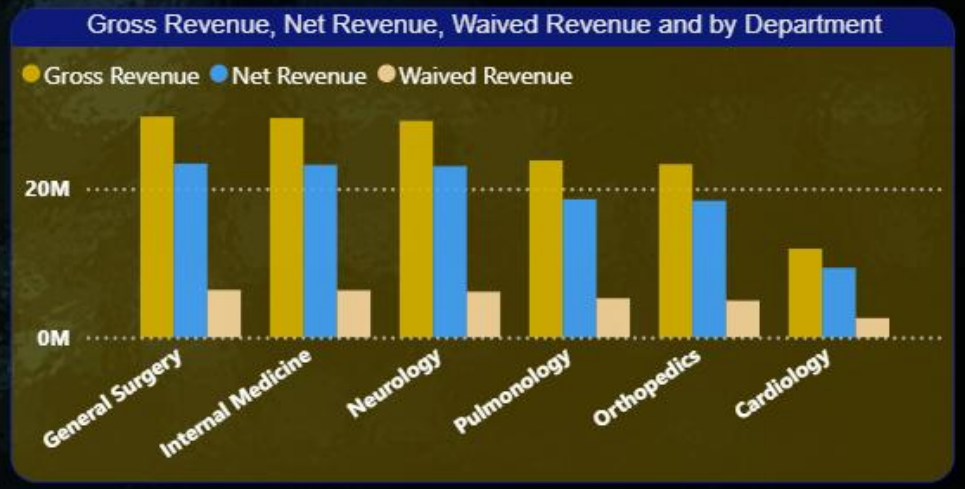
DASHBOARD – REVENUE CYCLE MANAGEMENT & FINANCIAL PERFORMANCE

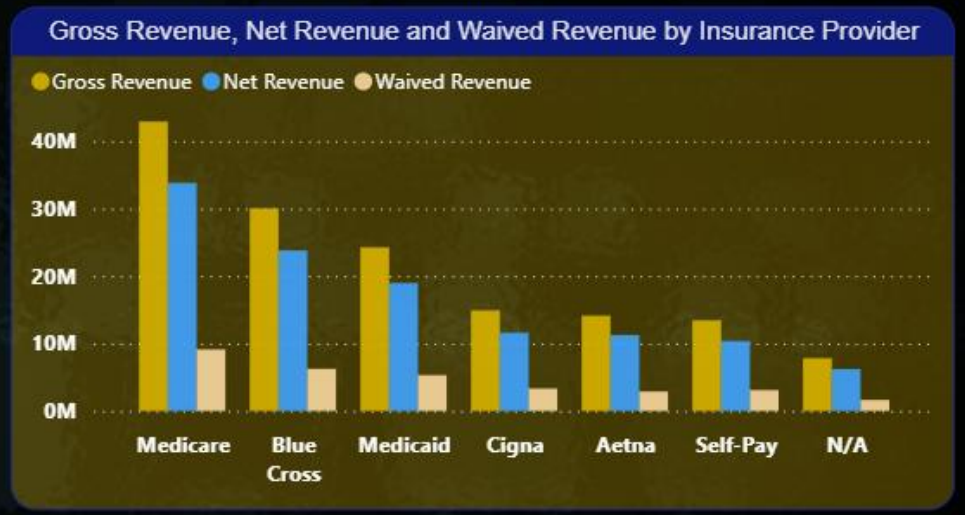
Home | RCM & Financial Performance | Patient Demographics | Clinical Quality & Satisfaction Ceiling

Gross Revenue	Net Revenue	Waived Revenue
147.38M	115.90M	31.49M

Net Conversion Rate	Total No. Of Bills	Total No. of Fully Waived Bills
78.64%	9000	40







Quarter

1 | 2 | 3 | 4

- Aetna
- Blue Cross
- Cigna
- Medicaid
- Medicare
- N/A
- Self-Pay

- Cardiology
- General Surgery
- Internal Medicine
- Neurology
- Orthopedics
- Pulmonology

- Elective
- Emergency
- Trauma
- Urgent

❑ RCM Interface Logic & Implementation Notes

Executive Focus Anchors: Positioned large card visuals prominently at the top-left to quickly orient leadership to the absolute fiscal targets: **$147.38M Gross Revenue**, **$115.90M Net Revenue**, a **78.64% Net Conversion Rate**, and **9,000 Total Bills**.

Leakage Quantification: Utilized a payer-level distribution bar chart to contrast revenue across insurance providers, successfully isolating **40 completely waived bills** and **$1.6M in unallocated cash leakage** hiding under missing provider fields ("N/A").

Chronological Tracking: Implemented a continuous dual-line visual model mapping monthly Gross vs. Net performance across the 2025 fiscal period to help executive leadership spot periodic contract compliance adjustments.

End-User Controls: Integrated an interactive right-margin slicer pane to support dynamic cross-filtering by Fiscal Quarter, Insurance Provider, Clinical Specialty, and Admission Type.

DASHBOARD – PATIENT DEMOGRAPHICS & PATIENT INTAKE INTEGRITY



❑ Demographics Interface Logic & Implementation Notes

Throughput Velocity Flux: Set up baseline high-level indicators (**25/Day Admissions** vs. **24/Day Discharges**) to give stakeholders an immediate look at structural hospital capacity.

Pipeline Disparity Isolation: Configured a department-level patient volume tracker that exposes critical operational gaps—showing high-volume areas like **General Surgery (1,812)** and **Internal Medicine (1,802)** contrasting against an underutilized **Cardiology department (726)**.

Acuity Mix Mapping: Deployed an admission type breakdown showing **Emergency encounters driving 52.56%** of total system volume, allowing clinical staffing teams to optimize resource placement based on case severity.

Demographic Segmentation: Applied a patient pipeline pie chart to parse gender-specific distributions, ensuring a clean visual structure to help non-technical users spot service-line trends.

DASHBOARD – CLINICAL QUALITY & SATISFACTION CEILING

Home | RCM & Financial Performance | Patient Demographics | Clinical Quality & Satisfaction Ceiling

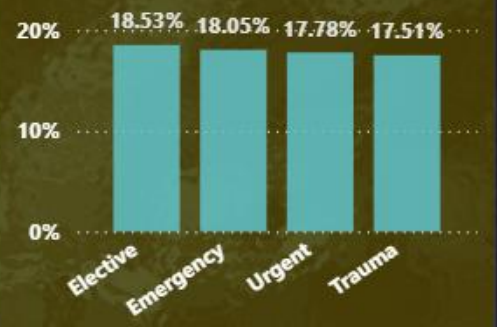
Avg Rating (out of 5)
3.61

Avg. Length of Stay (ALOS)
4.01 Days

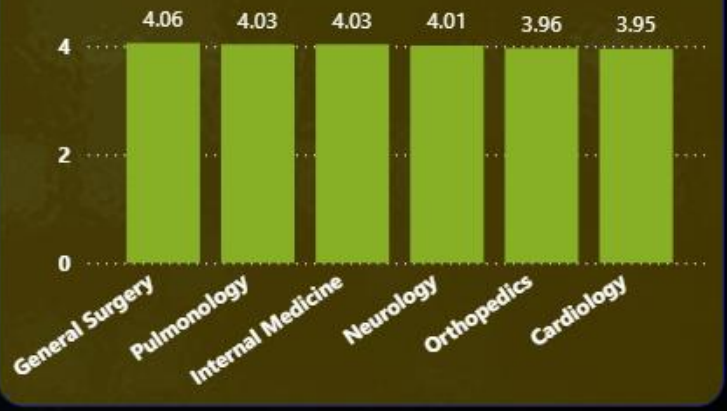
Readmission % (30 days)
18.14%

Total Readmitted Cases
1633

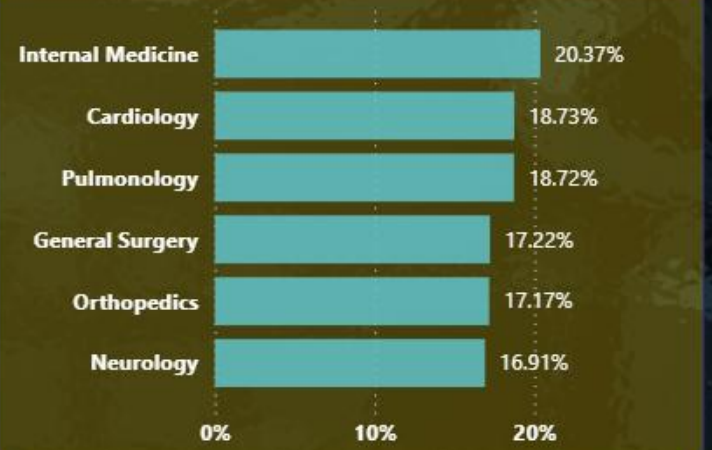
Readmission % (30 Days) by Admission Type



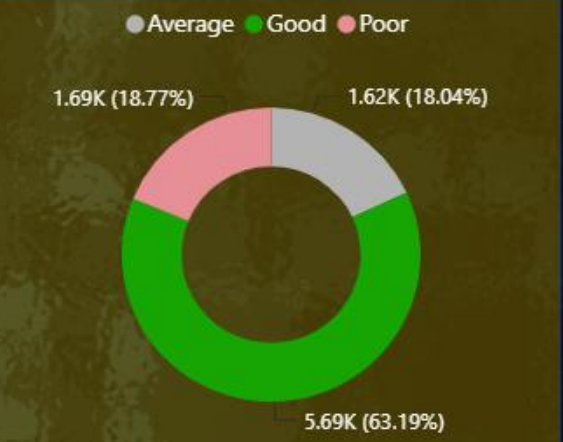
Avg Length of Stay (Days) by Department



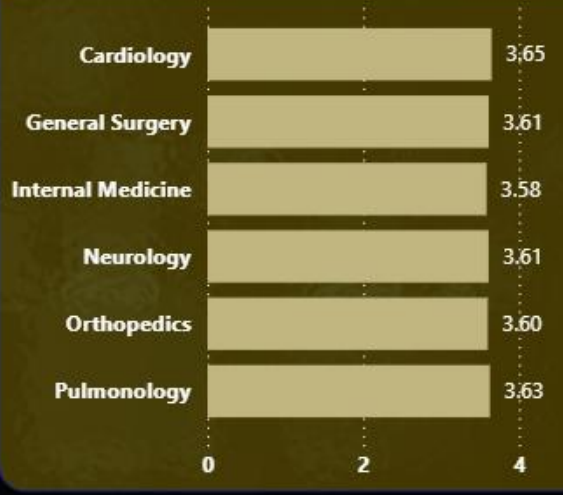
Readmission % (30 Days) by Department



Patient Satisfaction Breakdown



Avg Satisfaction Rating (out of 5) by Department



Quarter
1 | 2 | 3 | 4

Aetna
Blue Cross
Cigna
Medicaid
Medicare
N/A
Self-Pay

Cardiology
General Surgery
Internal Medicine
Neurology
Orthopedics
Pulmonology

Elective
Emergency
Trauma
Urgent

❑ Clinical Quality Interface Logic & Implementation Notes

Value-Based Care Risk Alerts: Highlighted penalty exposures under government quality guidelines by placing a high-impact warning card tracking a **30-day readmission rate of 18.14%** (representing **1,633 total readmitted cases**).

Stay Invariance Tracking: Visualized a flat **4.01 Day Average Length of Stay (ALOS)** across all primary medical categories to prove the presence of fixed system discharge constraints independent of diagnostic variations.

Targeted Risk Visuals: Engineered a horizontal bar chart to isolate department-level quality variations, pointing out that **Internal Medicine holds the highest readmission rate at 20.37%**.

Centralized Experience Engine: Structured a dynamic patient satisfaction breakdown (**63.19% Good** vs. **18.77% Average** and **18.04% Poor**) to demonstrate to stakeholders that patient experience limitations are consistent facility-wide.

EXECUTIVE SUMMARY

1. **Revenue Cycle & Registration Governance Risk**

Payer Integrity Disparities: An evaluation of the facility's commercial portfolio reveals that while the baseline contractual write-off rate ($31.48M or 21.3% of gross charges) matches expected industry benchmarks, significant system leakage is concentrated at the front-end intake tier.

Quantifiable Intake Leakage: Front-end data entry gaps allowed **185 individual patient encounters** to be processed under missing insurance categories ("N/A"), resulting in **$1.6M in completely unallocated cash write-offs**.

Contract Compliance Gaps: Commercial reimbursement auditing flags an active compliance issue within Cigna; despite holding an identical gross charge and demographic profile to top-performing commercial payers like Aetna, Cigna carries a net collection rate underperformance of nearly 2.0% (77.79% vs. 79.68%).

2. **Structural Specialty Line Capacity Under-Penetration**

Flat Service-Line Pricing: Cross-department pricing analytics show a flat average gross cost per case of approximately **$16.3K across all six medical specialties**, confirming that total departmental revenue variations are driven entirely by volume rather than pricing strategies.

Severe Service Line Bottlenecks: High-margin specialties suffer from severe regional market under-penetration; Cardiology operates at less than half the total consumer volume of General Surgery (**726 total admissions vs. 1,812 admissions**).

Physician Footprint Capacity: This volume deficit occurs despite massive operational headroom among the clinical team; the facility hospitalists currently average an active daily census of **only 4 patients per provider,** leaving significant structural capacity to scale up volume before hitting standard workload (12 to 15 cases per day).

3. Post-Discharge Care Continuity Deficits

Value-Based Care Exposures: The hospital faces heavy exposure to government financial penalties due to an elevated, cross-diagnostic **average 30-day readmission rate of 18.14%** (representing **1,633 total readmitted cases**).

Long-Stay Discharge Disconnects: While standard clinical readmission risks slope downward through Day 4 as acute symptoms stabilize, the network experiences a **severe spike back up to a 25.00% readmission rate** specifically for long-stay patient cohorts (12-day length of stay). This trend exposes an absolute absence of transitional post-acute tracking frameworks.

Systemic Experience Limitations: The hospital's patient satisfaction ceiling is flat at **3.61 out of 5 across all six medical departments**, and remains completely independent of whether a patient visits once or multiple times. This uniform mediocrity points to a centralized operational breakdown rather than individual provider performance.

STRATEGIC RECOMMENDATIONS & ACTION PLAN

1. **Institute Inpatient Transitional Care Management (TCM)**

Automated Care-Coordination Triggers: Deploy an automated alert within the electronic health record (EHR) system for all inpatient admissions that exceed a 5-day length of stay to systematically combat the documented **25.00% readmission risk spike observed at the Day 12 threshold**.

Post-Discharge Outreach: Mandate a strict clinical telephonic follow-up outreach framework within 48 hours of discharge, prioritizing high-risk conditions such as Urinary Tract Infections (21.92% readmission rate) and Heart Failure (21.91% readmission rate) to secure proper medication reconciliation and out-of-clinic stabilization

2. **Implement Front-End Insurance Verification Constraints**

Hard-Stop EHR Verification Requirements: Re-engineer the patient registration workflow to enforce a hard-stop eligibility validation requirement before booking non-emergent, elective admissions (which make up 30.17% of total volume).

Recovering Unallocated Revenue: Eliminating manual insurance processing oversights will immediately protect the facility's bottom line from the front-end registration gaps that allowed **$1.6M in gross revenue to slip into unassigned "N/A" write-off pools**.

3. **Launch Dynamic Specialty Line Marketing Campaigns**

Targeting Core Demographics: Leverage underutilized clinical capacity by launching regional outpatient marketing campaigns focused on chronic disease models and cardiovascular health lines specifically optimized for consumers **aged 36 and older, who command over 91% of total hospital volume**.

Absorbing High-Margin Volume: Utilizing the facility's existing medical staff capacity to capture an additional 1,000 cases in under-penetrated sectors like Cardiology (currently lagging at 726 admissions) can be executed smoothly **without expanding full-time employee (FTE) headcount or overhead**.

4. **Execute an Independent Commercial Payer Audit**

Re-negotiating Underperforming Portfolios: Mandate an immediate managed care compliance audit of the organization’s current fee schedule agreement with Cigna.

Standardizing Net Collections: Because Cigna's net collection rate (77.79%) structurally underperforms Aetna (79.68%) by approximately $500K for identical case-mix complexities, the reimbursement model must be forcefully re-aligned to match established local network pricing norms

THANK YOU!

By- Sanju